

GCSE Mathematics - Foundation Tier

Paper 3 (Calculator) Practice - Set 7

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use $\pi = 3.142$.

1.

Work out $\frac{1}{4}$ of 60.

(Total for Question 1 is 1 marks)

2.

A jumper costs £45 and is reduced by 30% in a sale. Work out the sale price.

(Total for Question 2 is 2 marks)

3.

(a) Round 7.39 to the nearest whole number.

(1)

(b) Round 234 to the nearest 10.

(1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 18 : 42 in its simplest form.

(Total for Question 4 is 2 marks)

5.

Two ordinary fair six-sided dice are rolled.

(a) How many possible outcomes are there in total?

(1)

(b) Work out the probability that the two dice show the same number.

(2)

(Total for Question 5 is 3 marks)

6.

Expand $5(3x - 2)$.

(Total for Question 6 is 2 marks)

7.

A rectangular field is 50 m long and 30 m wide. Sam walks from one corner of the field to the opposite corner in a straight line.

[Diagram: rectangle 50 m by 30 m with a diagonal drawn corner-to-corner.]

Work out the length of the diagonal walk. Give your answer to 1 decimal place.

(Total for Question 7 is 3 marks)

8.

The table shows the number of children per household for 20 households in a street.

Children	0	1	2	3	4
Freq.	4	6	7	2	1

Work out the mean number of children per household. Give your answer to 2 decimal places.

(Total for Question 8 is 3 marks)

9.

Town B is due east of Town A. Town C is on a bearing of 070 deg from Town A.

[Diagram: point A at origin; B is due east; C is in the NE quadrant such that the angle from North-line at A to AC measured clockwise is 70 deg.]

(a) Write down the bearing of B from A. (1)

(b) Work out the angle BAC. (2)

(Total for Question 9 is 3 marks)

10.

Solve $7x - 4 = 24$.

(Total for Question 10 is 3 marks)

11.

A cuboid has dimensions 12 cm by 5 cm by 4 cm.

(a) Work out the volume of the cuboid. (2)

(b) Work out the length of one of the longest edges. (You may state the answer.) (1)

(Total for Question 11 is 3 marks)

12.

Layla puts £5000 in a savings account paying 3.2% compound interest per year.

Work out the value of her investment after 2 years. Give your answer to the nearest penny.

(Total for Question 12 is 4 marks)

13.

A bag contains 8 sweets: 3 toffees and 5 mints. A sweet is taken at random, the colour noted, and then replaced. A second sweet is then taken.

- (a) Write down the probability of taking a toffee on a single pick. (1)
- (b) Work out the probability that both sweets taken are toffees. (3)

(Total for Question 13 is 4 marks)

14.

An isosceles triangle has a base of 16 cm and two equal sides of length 17 cm.

[Diagram: isosceles triangle with base 16 cm and two equal sides of 17 cm; perpendicular height from apex to base labelled h .]

- (a) Show that the perpendicular height h of the triangle is 15 cm. (3)
- (b) Hence work out the area of the triangle. (2)

(Total for Question 14 is 5 marks)

15.

A closed cuboid has dimensions 8 cm by 5 cm by 6 cm.

- (a) Work out the volume of the cuboid. (2)
- (b) Work out the total surface area of the cuboid. (3)

(Total for Question 15 is 5 marks)

16.

The n th term of a sequence is given by $T_n = 3n + 5$.

- (a) Write down the first three terms of the sequence. (2)
- (b) Work out T_{20} . (1)
- (c) Is 62 a term of the sequence? Show your working. (1)

(Total for Question 16 is 4 marks)

17.

The table shows the height (cm) and weight (kg) of 8 students.

Height	150	155	158	162	166	170	174	180
Weight	48	52	55	58	62	65	70	75

- (a) On the grid provided, plot these data as a scatter graph. (2)
- (b) Describe the correlation. (1)
- (c) Draw a line of best fit. (1)
- (d) Estimate the weight of a student who is 165 cm tall. (1)

(Total for Question 17 is 5 marks)

18.

A triangular prism has a cross-section that is a right-angled triangle with shorter sides of 6 cm and 8 cm. The length of the prism is 20 cm.

[Diagram: right-angled triangular prism: cross-section right-angled triangle with shorter sides 6 cm and 8 cm; length of prism 20 cm.]

- (a) Work out the volume of the prism. (3)

The prism is made of a metal of density 2.7 g/cm^3 .

- (b) Work out the mass of the prism. Give your answer in kilograms. (2)

(Total for Question 18 is 5 marks)

19.

A map of a country park uses a scale of 1 cm to 200 m.

- (a) Write the scale of the map in the form 1 : n. (2)
- (b) The distance between the visitor centre and the lake is 6.5 cm on the map. Work out the real distance in metres. (2)
- (c) The real distance between the lake and the play area is 1.4 km. Work out the distance on the map in centimetres. (1)

(Total for Question 19 is 5 marks)

20.

Asha and Ben share some money in the ratio 4 : 5. Ben receives £25 more than Asha. Let the amount Asha receives be £x.

- (a) Show that $x = 100$. (3)
- (b) Work out the total amount of money they share. (2)

(Total for Question 20 is 5 marks)

21.

Priya invests £8000 in a savings plan.

For the first 2 years it pays compound interest at 4% per year.

In year 3 the rate falls to 2% per year.

- (a) Work out the value of the investment at the end of year 2. (3)
- (b) Work out the value at the end of year 3. Give your answer to the nearest penny. (3)

(Total for Question 21 is 6 marks)

22.

60 students in a sixth-form chose one of three sports. The two-way table shows some information.

	Football	Hockey	Tennis	Total
Boys	16	7	—	30
Girls	—	9	13	—

Total _____ 60

- (a) Complete the two-way table. (3)
- (b) A student is chosen at random. Work out the probability that the student is a girl who chose Football. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 3 (Calculator) - Set 7 | 80 marks

Question 1

- 15 - B1

Total: 1 marks

Question 2

- Multiplier 0.7 or 30% of 45 = 13.50 seen - M1
- £31.50 - A1

Total: 2 marks

Question 3

- (a) 7 - B1
- (b) 230 - B1

Total: 2 marks

Question 4

- Both parts divided by HCF (e.g. 6:14 then 3:7) - M1
- 3 : 7 - A1

Total: 2 marks

Question 5

- (a) 36 - B1
- (b) Six "doubles" (1,1)(2,2)...(6,6) identified - M1
- (b) $6/36 = 1/6$ - A1

Total: 3 marks

Question 6

- First multiplication correct: $5 \times 3x = 15x$ or $5 \times (-2) = -10$ - M1
- $15x - 10$ - A1

Total: 2 marks

Question 7

- $d^2 = 50^2 + 30^2$ - M1
- $2500 + 900 = 3400$ - M1
- 58.3 (m) to 1 dp - A1

Total: 3 marks

Question 8

- $\text{Sum}(fx) = 0 + 6 + 14 + 6 + 4 = 30$ - M1
- $30 / 20$ - M1
- 1.50 - A1

Total: 3 marks

Question 9

- (a) 090 (deg) - B1
- (b) 90 - 70 attempted - M1
- (b) 20 (deg) - A1

Total: 3 marks

Question 10

- $7x = 28$ - M1
- $x = 4$ written - M1
- $x = 4$ with working clear - A1

Total: 3 marks

Question 11

- (a) $12 \times 5 \times 4$ attempted - M1
- (a) $240 \text{ (cm}^3\text{)}$ - A1
- (b) 12 (cm) - B1

Total: 3 marks

Question 12

- Multiplier 1.032 seen - M1
- $5000 \times 1.032 = 5160$ (year 1) - M1
- 5160×1.032 - M1
- £5325.12 - A1

Total: 4 marks

Question 13

- (a) $\frac{3}{8}$ - B1
- (b) Multiplication of independent events - M1
- (b) $\frac{3}{8} \times \frac{3}{8}$ - M1
- (b) $\frac{9}{64}$ - A1

Total: 4 marks

Question 14

- (a) Half the base = 8 cm identified - M1
- (a) $h^2 = 17^2 - 8^2 = 289 - 64 = 225$ - M1
- (a) $h = \sqrt{225} = 15 \text{ (cm)}$ shown - C1
- (b) Area = $0.5 \times 16 \times 15$ - M1
- (b) $120 \text{ (cm}^2\text{)}$ - A1

Total: 5 marks

Question 15

- (a) $8 \times 5 \times 6$ attempted - M1
- (a) $240 \text{ (cm}^3\text{)}$ - A1
- (b) $2(8 \times 5) + 2(8 \times 6) + 2(5 \times 6)$ - M1
- (b) $80 + 96 + 60$ - M1
- (b) $236 \text{ (cm}^2\text{)}$ - A1

Total: 5 marks

Question 16

- (a) Substitute $n = 1, 2, 3$ - M1
- (a) 8, 11, 14 - A1
- (b) $3 \times 20 + 5 = 65$ - B1
- (c) $3n + 5 = 62 \rightarrow n = 19$ (an integer) so YES 62 is a term - B1

Total: 4 marks

Question 17

- (a) All 8 points plotted correctly - B2 (B1 if 6+)
- (b) Positive correlation (strong) - B1
- (c) Sensible line of best fit - B1
- (d) Weight at 165 cm (accept 58 - 64 kg) - B1

Total: 5 marks

Question 18

- (a) $0.5 \times 6 \times 8 = 24$ (area of triangle) - M1
- (a) 24×20 - M1
- (a) $480 \text{ (cm}^3\text{)}$ - A1
- (b) $2.7 \times 480 (= 1296 \text{ g})$ - M1
- (b) 1.296 (kg) - A1

Total: 5 marks

Question 19

- (a) $200 \text{ m} = 20\,000 \text{ cm}$ - M1
- (a) $1 : 20\,000$ - A1
- (b) 6.5×200 - M1
- (b) 1300 (m) - A1
- (c) $1.4 \text{ km} = 1400 \text{ m}; / 200 = 7 \text{ cm}$ - B1

Total: 5 marks

Question 20

- (a) Asha = x , so one part = $x/4$ - M1
- (a) Ben = 5 parts = $5x/4 = x + 25$ - M1
- (a) $x/4 = 25 \rightarrow x = 100$ shown - C1
- (b) Asha £100, Ben = $5/4 \times 100 = £125$ - M1
- (b) Total = $100 + 125 = £225$ - A1

Total: 5 marks

Question 21

- (a) Multiplier 1.04 seen - M1
- (a) $8000 \times 1.04 = 8320$ (year 1) seen - M1
- (a) $8320 \times 1.04 = £8652.80$ - A1
- (b) Multiplier 1.02 used for year 3 - M1
- (b) 8652.80×1.02 - M1

- (b) £8825.86 (accept £8825.856 to 8825.86) - A1

Total: 6 marks

Question 22

- (a) Boys Tennis = $30 - 16 - 7 = 7$ - M1
- (a) Girls total = $60 - 30 = 30$; Girls Football = $30 - 9 - 13 = 8$ - M1
- (a) Column totals: Football 24, Hockey 16, Tennis 20 - A1
- (b) Girls Football = 8 out of 60 - M1
- (b) $8/60 = 2/15$ - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 3 (Calculator) - Set 7 | Detailed explanations

Question 1

$\frac{1}{4}$ means divide by 4.

$$60 \div 4 = 15.$$

Question 2

30% off means the sale price is 70% of the original.

$$0.7 \times 45 = 31.50.$$

Answer: £31.50.

Question 3

(a) 7.39: the next digit (tenths) is 3 (< 5), so round down. Answer: 7.

(b) 234: the units digit is 4 (< 5), so round down. Answer: 230.

Question 4

HCF of 18 and 42 is 6. Divide both parts: $18/6 = 3$, $42/6 = 7$.

Answer: 3 : 7.

Question 5

(a) Each die has 6 outcomes, so two dice give $6 \times 6 = 36$ ordered outcomes.

(b) "Same number" means doubles: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6). That is 6 outcomes.

$$P(\text{same number}) = 6 / 36 = 1/6.$$

Question 6

Multiply each term inside the bracket by 5.

$$5 \times 3x = 15x. \quad 5 \times (-2) = -10.$$

Answer: $15x - 10$.

Question 7

The diagonal is the hypotenuse of a right-angled triangle with legs 50 m and 30 m.

$$d^2 = 50^2 + 30^2 = 2500 + 900 = 3400.$$

$$d = \sqrt{3400} = 58.309..., \text{ i.e. } 58.3 \text{ m to 1 dp.}$$

Question 8

$$\text{Sum}(fx) = 0 \times 4 + 1 \times 6 + 2 \times 7 + 3 \times 2 + 4 \times 1 = 0 + 6 + 14 + 6 + 4 = 30.$$

$$\text{Sum}(f) = 20.$$

$$\text{Mean} = 30 / 20 = 1.50.$$

Question 9

- (a) Due east is 090° .
- (b) Angle BAC = bearing of B - bearing of C = $90 - 70 = 20^\circ$.
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Question 10

$7x - 4 = 24$. Add 4: $7x = 28$. Divide by 7: $x = 4$.
Check: $7(4) - 4 = 28 - 4 = 24$.

Question 11

- (a) $V = 12 \times 5 \times 4 = 240 \text{ cm}^3$.
- (b) The longest edge is the longest dimension - 12 cm.
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Question 12

Multiplier = 1.032.
Year 1: $5000 \times 1.032 = \text{£}5160.00$.
Year 2: $5160 \times 1.032 = \text{£}5325.12$.

Question 13

- (a) $P(\text{toffee}) = 3/8$.
- (b) After replacement, the second pick is independent of the first.
 $P(\text{both toffees}) = (3/8) \times (3/8) = 9/64$.
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Question 14

- (a) Drop the perpendicular from the apex to the base. It bisects the base, giving a right-angled triangle with hypotenuse 17 and one shorter side 8.
 $h^2 = 17^2 - 8^2 = 289 - 64 = 225$.
 $h = \sqrt{225} = 15 \text{ cm}$. (Shown.)
- (b) Area = $0.5 \times \text{base} \times \text{height} = 0.5 \times 16 \times 15 = 120 \text{ cm}^2$.
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Question 15

- (a) $V = 8 \times 5 \times 6 = 240 \text{ cm}^3$.
- (b) Three pairs of identical rectangular faces.
 $2(8 \times 5) + 2(8 \times 6) + 2(5 \times 6) = 80 + 96 + 60 = 236 \text{ cm}^2$.
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Question 16

- (a) $T_1 = 3(1) + 5 = 8$. $T_2 = 3(2) + 5 = 11$. $T_3 = 3(3) + 5 = 14$.
- (b) $T_{20} = 3(20) + 5 = 65$.
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(c) $3n + 5 = 62 \rightarrow 3n = 57 \rightarrow n = 19$. n is a whole number, so 62 is the 19th term. Yes.

Question 17

- (a) Plot 8 points.
 - (b) Positive correlation - taller students tend to be heavier.
 - (c) Best fit drawn.
 - (d) At 165 cm, line gives roughly 58 - 64 kg.
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Question 18

- (a) Cross-section is right-angled triangle: $\text{area} = 0.5 \times 6 \times 8 = 24 \text{ cm}^2$.
 $\text{Volume} = \text{area} \times \text{length} = 24 \times 20 = 480 \text{ cm}^3$.
 - (b) $\text{Mass} = \text{density} \times \text{volume} = 2.7 \times 480 = 1296 \text{ g}$.
Convert to kg: $1296 / 1000 = 1.296 \text{ kg}$.
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Question 19

- (a) 1 cm represents 200 m. Convert 200 m to cm: $200 \times 100 = 20\,000 \text{ cm}$. So the scale is 1 : 20 000.
 - (b) 6.5 cm on map = $6.5 \times 200 = 1300 \text{ m}$ in real life.
 - (c) 1.4 km = 1400 m. On map: $1400 / 200 = 7 \text{ cm}$.
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Question 20

- (a) Asha gets x (i.e. 4 parts), so one part = $x / 4$.
Ben gets 5 parts = $5 \times (x/4) = 5x/4$.
 $\text{Ben} - \text{Asha} = 5x/4 - x = (5x - 4x)/4 = x/4 = 25$.
So $x = 100$. (Shown.)
 - (b) Asha £100, Ben £125. Total = £225.
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Question 21

- (a) Years 1 and 2 multiplier = 1.04.
Year 1: $8000 \times 1.04 = \text{£}8320.00$.
Year 2: $8320 \times 1.04 = \text{£}8652.80$.
 - (b) Year 3 multiplier = 1.02.
Year 3: $8652.80 \times 1.02 = \text{£}8825.856$, i.e. £8825.86 to the nearest penny.
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Question 22

- (a) Boys row total 30, so Boys Tennis = $30 - 16 - 7 = 7$.
Girls row total = $60 - 30 = 30$, so Girls Football = $30 - 9 - 13 = 8$.
Column totals: Football $16+8=24$, Hockey $7+9=16$, Tennis $7+13=20$. Check: $24+16+20=60$.
 - (b) $P(\text{girl AND Football}) = 8/60 = 2/15$.
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